

Review

Overview of Chinese new energy vehicle industry and policy development

Jiamei Tian^a, Ping Wang^{a,*}, Daina Zhu^b^a School of International Economics and Trade, Xinjiang University of Finance & Economics, Xinjiang, 830012, China^b College of Education, Mississippi State University, Mississippi State, MS, 39762, USA

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ABSTRACT

The Chinese new energy vehicle (NEV) industry has developed rapidly, which has become one of the largest NEV markets in the world. The Chinese government has played a pivotal role in supporting and promoting the NEV industry, leading to significant advancements in policies, technology, infrastructure, industrial chain, and market development. This support has been evident through the implementation of numerous favorable policies, including increased support in finance, taxation, and technology innovation, as well as initiatives to promote research, development, and application of NEV products. Furthermore, subsidy policies have been introduced to incentivize consumers to purchase NEVs and enhance their competitiveness in the market. Additionally, regulations mandating a certain proportion of NEV production and sales for automakers have been put in place, contributing to the widespread advancement of the Chinese NEV industry. While the Chinese NEV industry has seen substantial growth, it also faces challenges and opportunities. To further its development, it is essential for vehicle manufacturers to prioritize technological innovation, the government to continue introducing supportive policies, users to increase environmental awareness, and for collaboration between academia and industry to drive research efforts. The development of the Chinese NEV industry is not only in line with the global trend of environmental protection, energy security, and industrial transformation, but also an important link in promoting the progress of the global NEV industry.

1. Introduction

Environmental pollution and the energy crisis are critical issues that the world is facing. Many countries are actively seeking and developing disposal measures. China has always been actively responding to the global call for environmental protection, which has put forward many environmental protection policies and measures. These measures include developing clean energy, promoting electric vehicles (EVs) (Zhang et al., 2022), and reducing coal use (Yuan, 2018). Replacing traditional fuel vehicles with EVs is a practical solution to the above issues (Onat and Kucukyar, 2022). The promotion of EVs has effectively reduced CO₂ emissions and oil consumption.

China has become one of the leaders in the global NEV market, which has taken multiple measures in response to the global promotion of NEVs (Wang et al., 2022). China has taken multiple measures to respond to the global promotion of NEVs, continuously strengthening its policies, technology, infrastructure, applications, and international cooperation, making positive contributions to the development of the global NEV market (Tian et al., 2022). The Chinese government has actively introduced various policies and standards to promote the development of the Chinese NEV

industry effectively (Tian et al., 2024a). These policies include car purchase subsidies, exemption from vehicle purchase tax, and free license plate acquisition to promote and popularize NEVs. The Chinese government has formulated environmental standards, including emission and energy efficiency standards for NEVs, to promote innovation and NEV technology development (Liu et al., 2023). The New Electric Vehicle Industry Plan lists new energy vehicles as one of China's strategic emerging industries and sets detailed plans and goals for the development of the NEV industry. (Wang et al., 2022). The government continues to increase infrastructure construction, invest in the construction of NEV charging infrastructure, promote the construction and operation of charging facilities, and improve the convenience and user experience of NEVs.

This work aims to provide a detailed overview and summary of the policies issued by the Chinese government in the NEV industry, involving both national and local levels. It summarizes the current development status of the Chinese NEV industry and qualitatively analyzes the impact of policies on industrial development.

The article is structured as follows: Section 2 elaborates on Chinese NEV policies at both national and local levels. Section 3 discusses the

* Corresponding author.

E-mail address: wangping@xjufe.edu.cn (P. Wang).<https://doi.org/10.1016/j.gerr.2024.100075>

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factors contributing to the growth of NEVs in the country. Section 4 analyzes the challenges and opportunities faced in the development of the Chinese NEV industry. Section 5 proposes measures to address these challenges and leverage opportunities. The conclusion is presented in Section 6.

2. Policy of EVs in China

The development history of electric vehicles in China can be traced back to the late 1990s. The following are the main events related to:

- > In 1995, China launched its first NEV (the “YuanWang”). It marks the first step taken by China in the field of NEVs..
- > From 2000 to 2005, China launched a modern automotive industry development plan, which conducted relevant research and trial production on electric vehicles, laying a technological foundation.
- > In March 2009, the General Office of the State Council of China released a plan for the adjustment and revitalization of the automotive industry, which pointed out that by implementing active consumption policies, expanding urban and rural markets, stabilizing and expanding the demand for automotive consumption (General Office of the State Council of China, 2009).
- > In 2012, Beijing, Shanghai, and other cities introduced purchase restrictions and license plate lottery measures to limit traditional fuel vehicles and promote EV sales (Quan and Xie, 2022).
- > In 2013, the Notice of the State Council on Issuing the Development Plan for Energy Conservation and New Energy Vehicle Industry (2012–2020) required the implementation of average fuel consumption management for passenger car enterprises, gradually reducing the average fuel consumption of China's passenger car products, and achieving the goal of reducing the average fuel consumption of China's passenger car products to 6.9 L/100 KM and 5.0 L/100 KM by 2015 and 2020 (Ministry of Industry and Information Technology of the People's Republic of China, et al., 2013).
- > In 2017, the Ministry of Industry and Information Technology, the Development and Reform Commission, and the Ministry of Science and Technology of China issued a notice on the Medium and Long Term Development Plan for the Automotive Industry. The document emphasized the vigorous development of advanced automotive technology, the formation of a reasonable industrial pattern, and a complete supporting system for the NEV industry (Ministry of Industry and Information Technology of the People's Republic of China, et al., 2017a).
- > In 2020, the State Council released the Development Plan for the New Energy Vehicle Industry (2021–2035), which focused on deepening the supply-side structural reform, adhering to the development direction of electrification, networking, and intelligence, breaking through key core technologies, enhancing industrial basic capabilities, building a new industrial ecosystem, and improving the infrastructure system (General Office of the State Council of China, 2020).
- > In 2023, the Chinese Ministry of Finance, the State Administration of Taxation, and other departments jointly issued a notice on continuing and optimizing the policy of reducing and exempting the purchase tax on NEVs. NEVs purchased between January 1, 2024, and December 31, 2025, are exempt from vehicle purchase tax. NEVs purchased between January 1, 2026, and December 31, 2027, are subject to a 50% reduction in vehicle purchase tax (Ministry of Finance of the People's Republic of China, 2023).

In summary, the development of Chinese NEVs has seen significant progress since the late 1990s. Key events such as the launch of China's first electric vehicle, the implementation of active consumption policies, the introduction of purchase restrictions for traditional fuel vehicles, and the issuance of development plans for the energy conservation and NEV industry have played crucial roles in shaping the

Chinese EV industry. Moreover, the continuation and optimization of policies, such as the reduction or exemption of the purchase tax for NEVs and the implementation of subsidy rules for vehicle trade-ins, demonstrate the government's commitment to promoting the popularity of Chinese electric vehicles and boosting the development of the NEV industry. Overall, the consistent efforts and support from the government have propelled the growth of the Chinese EV market, positioning the country as a key player in the global electric vehicle industry. The development process of NEV in China can be summarized as shown in Table 1.

2.1. National-level policies

In the past few years, the Chinese government has issued a large number of policies and plans for the NEV industry, including purchase subsidy policies, energy conservation and emission reduction policies (Wu et al., 2021), and supporting industrial policies for battery charging piles (Yang et al., 2013). These policies can be summarized in the following aspects:

- > NEV industry planning: In 2021, the “Development Plan for New Energy Vehicle Industry (2021–2035)” was released for proposing development goals and policy support measures in the NEV industry (National Development and Reform Commission, 2021).
- > Subsidy policy: Since 2010, the subsidy policy for NEVs has been implemented, which provides certain financial subsidies to eligible NEVs such as pure electric vehicles and plug-in hybrid vehicles. Starting in 2019, subsidy policies have gradually shifted towards fiscal incentive policies guided by technological innovation (Qu et al., 2022).
- > Indicator management policy: Since 2017, Sichuan and other provinces have implemented NEV indicator management policies to limit the number of conventional fuel vehicles and encourage the use of NVEs (The People's Government of Sichuan Province, 2017).
- > Energy consumption double point policy: The “Dual Points” management measures were released in September 2017, aiming to improve the energy efficiency level of passenger cars, establish a long-term mechanism for energy conservation and emission reduction of passenger cars, and manage NEVs, which is conducive to promoting the healthy development of the automotive industry (Ministry of Industry and Information Technology of the People's Republic of China, et al., 2017b).
- > Charging infrastructure construction: The government has increased investment in the construction of charging infrastructure for NEVs, supported the construction of charging stations and encouraged the participation of social capital (General Office of the State Council of China, 2023).
- > Promotion and application of NEVs: The State promotes the application of NEVs in government procurement, public transport, urban logistics, taxis and other fields, which is conducive to the promotion of NEV market development (Ministry of Transport of the People's Republic of China, 2015).

Table 2 shows the policies related to the NEV industry at the national level. The publishing agencies mainly involve the National Development and Reform Commission, the State Council, the Ministry of Finance of the People's Republic of China, etc. As can be seen from the main content of policies, these policies are mainly divided into supportive and normative types. The former advocates increasing purchase subsidies, reducing vehicle and vessel insurance, or accelerating the construction of charging infrastructure (Ma et al., 2019). The latter specifies vehicle testing and charging standards (Park et al., 2022). These policies benefit automakers, third-party testing agencies, and consumers, further promoting the production and sales of NEVs in China.

Table 1
Development history of NEVs in China.

Stage	Year	Main policies	Promotion scope	Category
◇Demonstration	2009	✓ Notice on Pilot Work of Energy Saving and New Energy Vehicle Demonstration and Promotion	13 pilot cities (Beijing, Shanghai, Chongqing, Changchun, Dalian, Hangzhou, Jinan, etc.)	EVs for public services
	2010–2012	✓ Notice on Carrying out the Pilot Program of Subsidies for Private Purchase of New Energy Vehicles	5 pilot cities (Shanghai, Changchun, Shenzhen, Hangzhou, Hefei)	Private EVs
◇Promotion & application	2013–2017	✓ Notice on Continuing the Promotion and Application of New Energy Vehicles	88 pilot cities (Beijing, Shanghai, Changchun, Nanjing, etc.)	Various NEVs
		✓ Notice on Further Improving the Promotion and Application of New Energy Vehicles		
		✓ Notice on Adjusting the Financial Subsidy Policy for the Promotion and Application of New Energy Vehicles		
◇Marketization adjustment	2018–2023	✓ Notice on Adjusting and Improving the Financial Subsidy Policy for the Promotion and Application of New Energy Vehicles	Nationwide	Various NEVs
		✓ Notice on Further Improving the Financial Reform Subsidy Policy for the Promotion and Application of New Energy Vehicles		
		✓ Notice on the Financial Subsidy Policy for the Promotion and Application of New Energy Vehicles in 2022		

Table 2
NEV policies issued at the national level.

Date	Issuing agencies	Policies	Primary coverages
2023.12.13	◇ National Development and Reform Commission ◇ National Energy Administration ◇ Ministry of Industry and Information Technology of the People's Republic of China ◇ State Administration for Market Regulation	Implementation Opinions on Strengthening the Integration and Interaction between New Energy Vehicles and the Power Grid	✓ Collaborative promotion of core technologies for vehicle network interaction ✓ Accelerate the establishment of a standard system for car network interaction ✓ Optimize and improve supporting electricity prices and market mechanisms ✓ Explore the comprehensive demonstration of bidirectional charging and discharging ✓ Actively improving the interaction level of charging and swapping facilities ✓ Systematically strengthening the support and guarantee capabilities of power grid enterprises
2023.06.08	◇ State Council	Guiding Opinions of the General Office of the State Council on Further Building a High-Quality Charging Infrastructure System	✓ Build a high-quality charging infrastructure system with extensive coverage and complete functions ✓ Construct a charging network that forms a layout of urban areas, highway lines, and rural areas ✓ Increased coverage of charging services in rural areas
2023.05.14	◇ National Energy Administration	Implementation Opinions on Accelerating the Construction of Charging Infrastructure and Better Supporting the Rural Revitalization of New Energy Vehicles	✓ Support the purchase and use of new energy vehicles in rural areas and provide support such as consumption vouchers ✓ Encourage relevant car companies and places with conditions to provide rewards for exchanging old cars for new ones ✓ Encourage local governments to strengthen the linkage between government and enterprises ✓ Increase support for automobile consumption credit in rural areas
2023.01.30	◇ Ministry of Industry and Information Technology of the People's Republic of China ◇ National Development and Reform Commission ◇ Ministry of Finance of the People's Republic of China ◇ Ministry of Ecology and Environment of the People's Republic of China ◇ Ministry of Housing and Urban-Rural Development of the People's Republic of China ◇ National Energy Administration ◇ State Post Bureau of The People's Republic of China	Notice on Organizing Pilot Work for Comprehensive Electrification of Public Sector Vehicles in Pilot Areas	✓ Improve the level of vehicle electrification, striving to reach 80% in the fields of urban public transportation, rental, environmental sanitation, postal and express delivery, and urban logistics distribution ✓ Enhance the guarantee capability of charging and swapping services to ensure a 1:1 ratio between the number of new public charging stations and the number of NEVs promoted in the public sector ✓ Build a batch of typical comprehensive energy service demonstration stations

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Table 2 (continued)

Date	Issuing agencies	Policies	Primary coverages
2022.09.18	◇Ministry of Finance of the People's Republic of China	Announcement on Extending the Policy of Exempting New Energy Vehicles from Vehicle Purchase Tax	✓ NEVs purchased from January 1, 2023, to December 31, 2023, are exempt from vehicle purchase tax
2022.01.10	◇State Taxation Administration ◇National Development and Reform Commission	Implementation Opinions on Further Improving the Service Guarantee Capacity of Electric Vehicle Charging Infrastructure	✓ Explore the implementation path for NEVs to participate in the electricity spot market ✓ Study the trading and scheduling mechanisms for NEVs to consume and store green electricity
2021.12.28	◇State Council	Notice on the Comprehensive Work Plan for Energy Conservation and Emission Reduction for the 14th Five-year Plan Period	✓ Increase the proportion of NEVs used by buses, taxis, etc. ✓ Implement national emission standards for automobiles
2021.10.24	◇State Council	Action Plan for Carbon Peak Before 2030	✓ Promote low-carbon transformation of transportation equipment ✓ Reduce the ownership of traditional fuel vehicles ✓ Promote the electrification of urban public service vehicles
2021.08.23	◇State Administration for Market Regulation	Announcement on Regulating the Testing and Charging of New Energy Vehicles	✓ Vehicle testing agencies set up testing fee items ✓ List the service content and mark the price
2021.02.22	◇State Council	Guiding Opinions on Accelerating the Establishment and Improvement of a Green and Low Carbon Circular Development Economic System	✓ Actively build green highways and railways ✓ Strengthen the construction of NEV charging and switching infrastructure
2020.04.16	◇Ministry of Finance of the People's Republic of China ◇State Taxation Administration ◇Ministry of Industry and Information Technology of the People's Republic of China	Announcement on the Policy of Exempting New Energy Vehicles from Vehicle Purchase Tax	✓ Exemption from vehicle purchase tax
2019.05.08	◇Ministry of Finance of the People's Republic of China ◇Ministry of Industry and Information Technology of the People's Republic of China ◇Ministry of Transport of the People's Republic of China ◇National Development and Reform Commission	Notice on Supporting the Promotion and Application of New Energy Buses	✓ Local governments can continue to provide subsidies and support for the purchase of new energy buses ✓ Implement the policy of exemption from vehicle purchase tax on new energy buses
2018.01.09	◇National Development and Reform Commission ◇National Energy Administration ◇Ministry of Industry and Information Technology of the People's Republic of China ◇Ministry of Finance of the People's Republic of China	Action Plan for Improving the Charging Support Capacity of New Energy Vehicles	✓ Improve charging technology level ✓ Improve product quality of charging facilities ✓ Accelerate the improvement of the charging standard system ✓ Fully optimize the layout of charging facilities ✓ Significantly enhance charging network connectivity
2017.04.06	◇Ministry of Finance of the People's Republic of China ◇National Development and Reform Commission ◇Ministry of Science and Technology of the People's Republic of China	Medium and Long-Term Development Plan for the Automobile Industry	✓ Accelerate the R&D and industrialization of NEV technology ✓ Utilize social capital invested by enterprises and the national science and technology plan to coordinate and organize collaborative research among enterprises, universities, scientific research institutes, etc.
2015.05.08	◇State Council	Made in China (2025)	✓ Continue to support the development of electric vehicles ✓ Improve the engineering and industrialization capabilities of core technologies such as power batteries and drive motors

2.2. Local-level policies

Under the advocacy of national policies, local provinces and cities in China are also continuously launching NEV policy plans. These policies mainly involve the expected goals of output value, supporting industrial development, and other aspects. Table 3 shows the crucial policies on NEVs introduced by major provinces in 2022 and 2023.

NEV policies issued by the Chinese central government influence the implementation of local policies in provinces and cities. National policies set the framework system for the development and implementation of local policies. Local governments need to consider the unique circumstances of their respective regions and formulate a series of measures suitable for central policies based on the actual situation to promote the development of the NEV industry. In summary, national policies have played a guiding role in the implementation and adjustment of local policies in various provinces and cities. They may also be influenced by specific situations in different regions, resulting in certain differences. The impact of the NEV policies issued by the central government of China

on local policies in various provinces and cities is mainly manifested in the following aspects:

- Promote the development of the local NEV industry ([National Development and Reform Commission, 2021](#)): National-level policies guide local governments to increase their support for the NEV industry and promote its development.
- Unify market access standards for NEVs ([Ministry of Industry and Information Technology of the People's Republic of China, 2022](#)). National-level policies provide local governments with unified market access and technical standards, avoid differences between local policies, and promote the standardization and healthy development of the NEV market.
- Promotion and application of NEVs ([Ministry of Transport of the People's Republic of China, 2015](#)): Local governments promote the promotion and application of NEVs in public transportation, urban logistics, and encourage local governments to take corresponding preferential measures.

Table 3
NEV policies in mainland China from 2022 to 2023.

Provinces	Data	Policies
Guangdong	2022.04.29	Announcement on New Energy Vehicle Purchase Subsidy Activities in Guangdong Province
	2023.11.24	Several Measures for Guangdong Province to Further Boost and Expand Consumption
Jiangxi	2022.05.10	Activity Plan for Jiangxi Bulk Commodity Consumption Season in 2022
	2023.07.10	Notice on Energy Conservation and Emission Reduction Subsidy Funds (Second Batch) for 2023
Shandong	2022.05.22	Several Measures to Promote Automobile Consumption
	2023.12.06	White Paper on the Development of Electric Vehicle Charging Infrastructure in Shandong Province
Henan	2022.05.11	Guiding Opinions on Accelerating the Development of New Energy Vehicle Industry
	2023.08.07	Three-year Action Plan for the Construction of Electric Vehicle Charging Infrastructure in Henan Province (2023–2025)
Hubei	2022.05.01	Several Measures for Accelerating the Recovery of Consumption
	2023.03.17	Implementation Plan for the Construction and Transformation of New Energy Vehicle Charging Infrastructure in Hubei Province in 2023
Hunan	2022.06.14	Several Policy Measures to Support the High-quality Development of the New Energy Vehicle Industry
	2023.11.30	Electric Vehicle Charging Infrastructure Construction and Operation Management Measures in Hunan Province
Hainan	2022.03.29	Several Measures to Encourage the Use of NEVs in Hainan Province in 2022
	2023.09.24	Several Measures to Encourage the Promotion and Application of New Energy Vehicles in Hainan Province from 2023 to 2025
Hebei	2022.08.10	Guidelines for Application of NEV Industry Development Projects in 2022
	2023.09.20	Several Measures to Promote Automobile Consumption in Hebei Province
Shanxi	2022.12.31	2022 Work Plan for Shanxi Province to Support the Development of New Energy Industry
	2023.07.30	Three-year Action Plan for the Construction of Electric Vehicle Charging (Swapping) Infrastructure in Shanxi Province (2023–2025)
Liaoning	2022.03.09	Notice on Carrying out the Application of Financial Subsidy Fund for the Development of New Energy Vehicle Industry in 2022
	2023.09.19	Implementation Opinions on Accelerating the Construction of Charging Infrastructure in the Transportation Sector and Promoting the Application of New Energy Vehicles
Jilin	2022.11.11	Several Measures to Further Promote Automobile Consumption in Jilin Province
	2023.11.07	Notice on Improving Electricity Prices and Service Fees for Electric Vehicles
Heilongjiang	2022.12.31	Measures to enliven the circulation of automobiles and expand the upgrading of automobile consumption
	2023.12.21	Catalogue of New Energy Vehicle Models Exempted from Vehicle Purchase Tax
Jiangsu	2022.03.21	Administrative Measures for the Construction and Operation of New Energy Vehicle Charging (Replacement) Facilities in Jiangsu Province
	2023.04.06	Implementation Plan for Jiangsu Province's Electric Vehicle Charging (Swapping) Infrastructure Planning (2023–2025)
Zhejiang	2022.06.08	Notice on Issuing the Budget for Energy Conservation and Emission Reduction Subsidy Fund (New Energy Vehicle Charging Infrastructure Award) from the Central Government in 2022
	2023.10.20	Action Plan for Improving the Network System of High-Quality Charging Infrastructure and Promoting the Rural Development of New Energy Vehicles (2023–2025)
Anhui	2022.08.11	Several policies to support the improvement of quality, expansion, and efficiency in the New Energy Vehicle and intelligent networked vehicle industries
	2023.12.28	Regulations on the Development of New Energy Vehicle Industry Clusters in Anhui Province
Fujian	2022.04.22	Fujian New Energy Vehicle Industry Development Plan (2022–2025)
	2023.06.12	Implementation Opinions on Comprehensively Promoting the Construction of “Electric Fujian” (2023–2025)
Sichuan	2022.03.30	“Electric Sichuan” Action Plan (2022–2025)
	2023.11.03	Sichuan Province Accelerates the Construction of Charging Infrastructure to Support New Energy Vehicles Going to Countryside and Rural Revitalization Work Plan
Guizhou	2022.03.07	Measures to enliven automobile circulation and expand automobile consumption in Guizhou Province
	2023.11.03	Guiding Opinions on Accelerating High-Quality Development of New Energy Vehicle Industry and Promoting the Construction of “Electric Guizhou”
Yunnan	2022.07.15	Activities of New Energy Vehicles in the Countryside in Yunnan Province in 2022
	2023.01.18	Notice on Issuing Policies and Measures to Promote Economic Stability and Quality Improvement in 2023
Shaanxi	2022.11.30	Three-year Action Plan for Accelerating the Construction of New Infrastructure in Shaanxi Province (2022–2024)
	2023.08.11	Special Work Plan for Comprehensive Electrification of Public Vehicles in Shaanxi Province (2023–2027)
Gansu	2022.05.09	Implementation Opinions on the Development of New Energy Vehicle Industry in Gansu Province
	2023.12.22	Several Implementation Measures of Gansu Province on Promoting Automobile Consumption
Beijing	2022.03.14	Beijing's Urban Management Development Plan during the “14th Five-year Plan” Period
	2023.10.20	Implementation of Rules for Rewards for the Construction and Operation of Electric Vehicle Charging and Exchange Facilities in Beijing in 2023
Shanghai	2022.10.01	Measures of Shanghai Municipality on Encouraging the Development of Electric Vehicle Charging and Switching Facilities
	2023.12.14	Implementation Measures of Shanghai Municipality on Encouraging the Purchase and Use of New Energy Vehicles
Tianjin	2022.03.29	Implementation Plan for Tianjin to Accelerate the Establishment and Improvement of a Green and Low Carbon Cyclic Development Economic System
	2023.09.28	Implementation Plan for Accelerating the Development of New Energy and Intelligent Connected Vehicle Industry in Tianjin (2023–2027)
Chongqing	2022.09.28	Chongqing Action Plan for Promoting Infrastructure Construction and Service of Intelligent Connected New Energy Vehicles (2022–2025)
	2023.11.30	Special Action Plan for Chongqing Intelligent Connected NEV Parts Industry Cluster Enhancement (2023–2027)
Neimenggu	2022.02.20	List of Policies for Inner Mongolia Autonomous Region to Adhere to Steady Progress and Promote High-quality Industrial Development in 2022
	2023.04.06	Six Policy Measures to Promote the Construction of Urban NEV Charging Facilities
Guangxi	2022.04.25	Guangxi Zhenzuo Industrial Economic Operation Promotes High-quality Industrial Development 2022 Key Tasks List
	2023.08.20	Guangxi accelerates the construction of charging infrastructure.
Ningxia	2022.03.22	Ningxia Charging Infrastructure “14th Five-year Plan”
	2023.11.01	Several Policy Measures for Restoring and Expanding Consumption
Xinjiang	2022.09.17	“Several Measures of the Autonomous Region to Promote the Recovery of Automobile Consumption”
	2023.05.05	Implementation Plan for the Restoration and Expansion of Consumption in the Autonomous Region in 2023
Xizang	2022.07.05	Measures to Revitalize the Circulation of Automobiles and Expand Automobile Consumption
	2023.03.24	Xizang Autonomous Region's “Happy Xizang, Joyful Consumption” Special Action Plan to Boost Consumption

In June 2021, the China Association of Automobile Manufacturers unveiled “14th Five-year Plan for the Development of the Automotive Industry”, which highlighted the successful establishment of a new industrial ecosystem focusing on the NEV industry. Breakthroughs have been made in overcoming common technological bottlenecks, leading to brands with strong international competitiveness. The infrastructure system has been continuously improved, thereby increasing the affordability and convenience of products. The improvement in the market environment has led to a significant increase in product penetration rate (China Association of Automobile Manufacturers, 2021). Fig. 1 summarizes the development goals of the NEV industry in China's provinces and cities during the 14th Five-year Plan.

3. Analysis of NEVs industry development

The rapid development of the Chinese NEV industry has benefited from both favorable policy support and technological innovation. Specifically, it can be summarized as follows:

3.1. Technological innovation

Chinese NEVs have significantly progressed in core technologies, including batteries, motors, and electronic controls (Du et al., 2018.). Companies such as Contemporary Amperex Technology Co. and BYD occupy a leading position in the global power battery market. In 2023, Chinese NEV companies are expected to continue to increase research and development investment, promote technological innovation, and improve product performance and competitiveness, such as battery, electronic control, and vehicle networking technology.

Lithium-ion battery technology continues to advance, including improvements in materials, reductions in production costs and the design of high-performance structural systems. These advanced technologies have greatly improved the range and performance of NEVs. According to SNE Research statistics, the global installed capacity of power batteries reached 705.5 GWh in 2023. Fig. 2 summarizes the top 10 global power

battery installation companies (My B. 2023). Chinese companies occupy six seats, with a market share of 63.5%, an increase compared to 59.6% in 2022. The combined market share of CATL and BYD is over half, reaching over 50%.

The rapid growth in the installed capacity of power lithium batteries is also due to the decrease in battery prices. Since 2018, the overall price of power lithium batteries in China has shown a downward trend. The average price of power lithium battery cells has decreased from 0.75 RMB/Wh in 2017 to 0.52 RMB/Wh in 2021, as shown in Fig. 3 (Forward, L., 2024.). In 2022, due to a significant increase in upstream material prices, the average price of power lithium battery cells has increased significantly to 0.79 RMB/Wh. In 2023, the price of power lithium battery materials has dropped, with the average price of battery cells around 0.55 RMB/Wh. The reasons for this phenomenon can be summarized as follows.

The imbalance between supply and demand in China's power lithium battery industry has led to a price war among power lithium battery companies, resulting in a decline in prices.

Since 2023, the prices of key materials for power batteries in China have decreased, driven by a reduction in production costs. According to the 2023 Study on the Full Life Cycle Cost of Lithium Battery New Energy Vehicles, in the cost composition of power lithium battery cells in China, positive electrode materials, separators, electrolytes, and negative electrode materials account for approximately 40%, 25~30%, 15%~20%, and 10% of the total cost of battery cells, respectively (Forward, L., 2024.). In 2023, the prices of power lithium battery materials have all experienced varying degrees of decline, with positive electrode materials experiencing the largest decline. From January to December, the price of the LiFePO₄ power type decreased by 69.5%, and the price of the ternary materials power type decreased by 58.5%. The comprehensive decrease in the price of power lithium-ion battery materials has led to a significant decrease in the production cost of power lithium-ion batteries, thereby driving down the price of power lithium-ion batteries.

Moreover, the continuous decline in prices of NEVs has squeezed the profit margins of manufacturers. In order to alleviate business pressure, manufacturers have lowered the purchase price of power batteries.

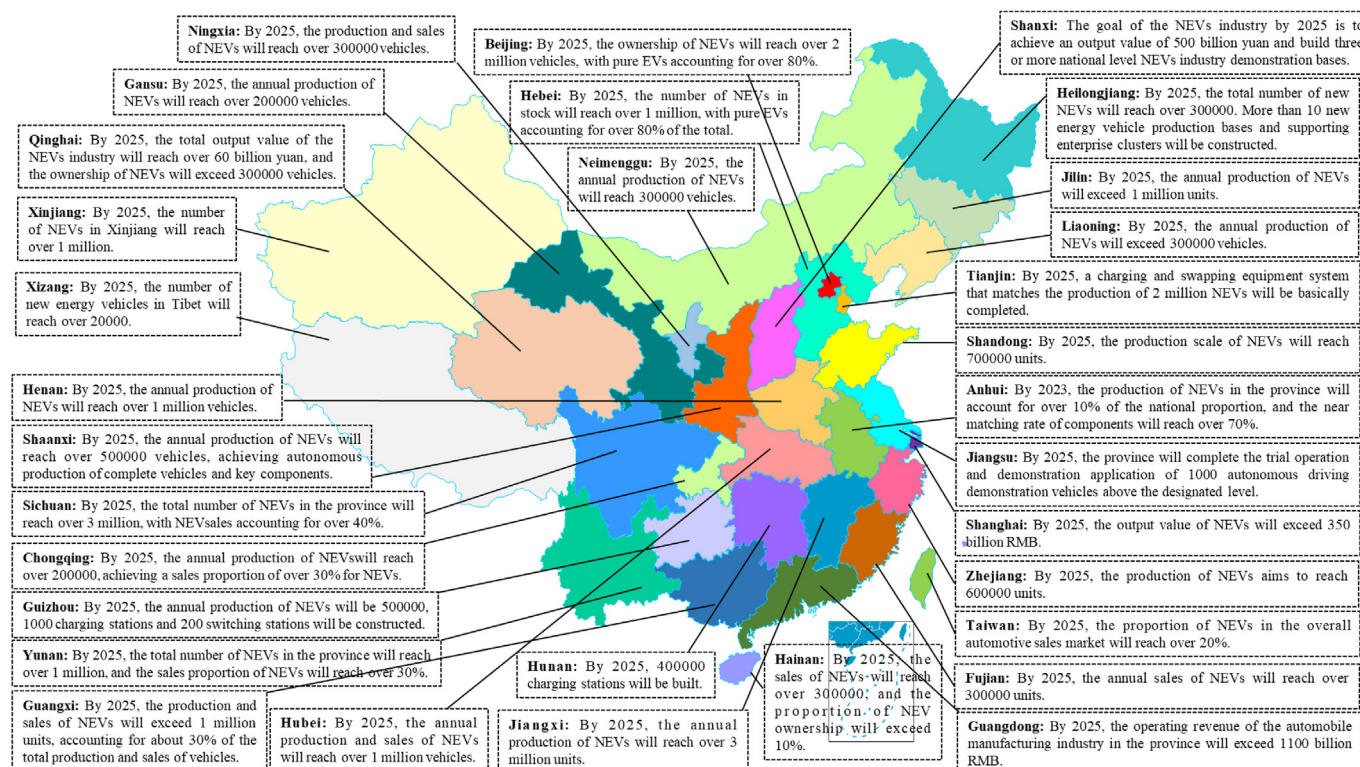


Fig. 1. The planning goals of the “14th Five-year Plan for NEV Industry” in various provinces and cities in China.

The continuous development of electric drive technology, such as the expansion of the motor speed range and the improvement of motor power density, has made the power system of NEVs more efficient and reliable while reducing noise and emissions (Higuchi et al., 2013). The manufacturing process of motor components is becoming more and more mature, while the cost of materials is also falling. This has made the manufacturing cost of electric vehicles more competitive, driving the rapid growth of the new energy vehicle market (Cai et al., 2021).

3.2. Charging infrastructure construction

Intelligent charging technology can effectively solve the charging speed of NEVs, which is beneficial for improving user convenience and service quality (Li et al., 2022). As of the end of 2021, the number of charging stations in China exceeded 2 million, a year-on-year increase of about 50%. In 2022, the increment of charging infrastructure was 2.593 million units, and the increment of public charging stations increased by 91.6% year-on-year. It is expected that in 2024, the government will continue to increase its support for the construction of charging infrastructure, and the coverage of charging networks will be further expanded, providing strong support for the development of NEVs.

3.3. Industrial chain improvement

The Chinese NEV industry chain is gradually improving. Sales and services have evolved from raw material and component manufacturing to vehicle manufacturing (Zhao, 2023). Each link has specialized enterprises for production and supply. In recent years, the upstream and downstream enterprises in China's new energy vehicle industry chain have become more prominent, continuously segmenting fields, and establishing a business model of vertical integration and collaborative development. The industrial chain continues to progress and develop, and competition and cooperation between enterprises drive the optimization and upgrading of the industrial chain.

3.4. Diversified competitive landscape

The Chinese NEV industry is gradually becoming one of the world's leading industries. Multiple companies are competing fiercely in the Chinese NEV market, including BYD, NIO, XPENG, LEADING IDEAL, and others (Liu et al., 2022). These enterprises have muscular strength and development advantages in the research, development, production, and sales of NEVs. From the perspective of vehicle models, the Chinese NEV market is mainly divided into electric and plug-in hybrid vehicles. Among them, electric vehicles are powered by pure electricity, while companies such as NIO and XPENG are essential participants in the electric vehicle market. On the other hand, plug-in hybrid vehicles

combine the advantages of traditional engines and electric motors and are developed and produced by companies such as BYD and LEADING IDEAL (Jia et al., 2023). Overall, the competitive landscape of the Chinese NEV industry is very complex, with many different enterprises competing. It also indicates the enormous potential of the Chinese NEV market, with broad development prospects and market opportunities.

In summary, the development prospects of China's new energy vehicle industry are broad in 2023. Policy support, technological innovation, and market demand jointly promote the rapid development of the industry. However, the industry is facing challenges such as intensified competition, technological breakthroughs, and charging infrastructure construction, which require continuous attention and response.

4. Challenges and opportunities

4.1. Challenge from competitors

With the rapid growth of the global NEV market, the Chinese NEV market is also facing fierce competition. It has evolved into one of the world's largest, drawing significant interest from both domestic and foreign automobile manufacturers (Cao et al., 2022). These competitors enjoy distinct advantages in technology research and development, capacity expansion, and other areas, which are driving substantial impact and posing challenges for the Chinese NEV industry. The competitive landscape can be categorized into several main aspects: technology and quality competition, brand building and market competition (Dong and Zhang, 2022).

4.1.1. Technology and quality competition

In the realm of NEV technology, battery technology lies at the heart of core competencies. Diverse battery technologies can impact crucial factors such as the range, safety performance, and overall driving experience of NEVs. In addition, consumers place a strong emphasis on quality issues when deciding which NEV to purchase (Fujita et al., 2022).

> Technological innovation: Both domestic and foreign competitors are actively engaged in technological innovation to enhance the performance and quality of their products. Tesla's remarkable achievements in battery technology, autonomous driving, and intelligence have posed a significant challenge for Chinese NEV manufacturers (Yang et al., 2023). To tackle this challenge, manufacturers must prioritize the reinforcement of their technological research and development efforts while enhancing the technological content and overall competitiveness of their products. This can be achieved through increased investment in research and development, strategic partnerships with technologists, and continuous improvement of manufacturing processes (Han et al.,

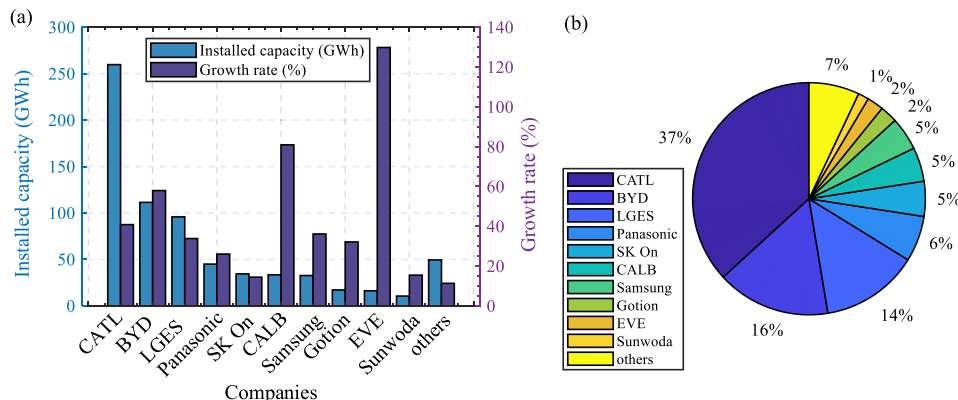


Fig. 2. Top 10 global power battery installations in 2023. (a) Global power lithium battery installation statistics. (b) Market share in 2023.

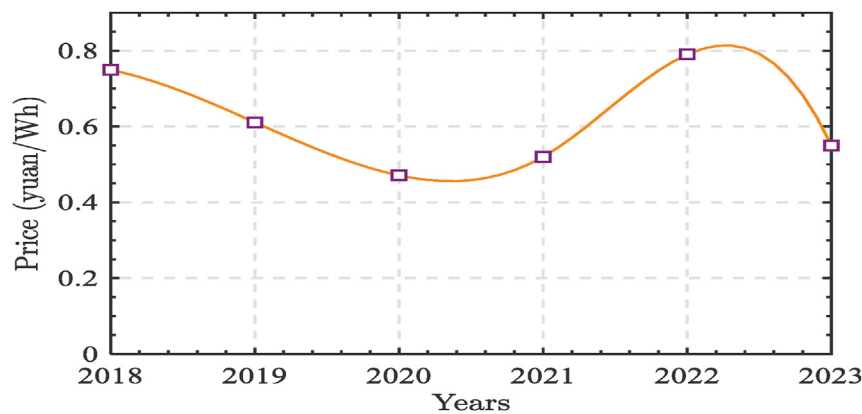


Fig. 3. The average price trend of power battery cells in China from 2017 to 2023.

2022). Manufacturers can effectively meet customer expectations and maintain a strong position in the marketplace by staying at the forefront of technological advancement.

- **Quality improvement:** Consumers have been increasingly demanding in terms of the quality and reliability of NEVs. As a result, Chinese manufacturers are driven by a competitive market to continuously enhance their manufacturing processes and quality management systems, aiming to deliver more reliable and high-quality NEV products (Ma et al., 2019).

4.1.2. Brand building and market competition

Brand building and market competition play vital roles in the development of the Chinese NEV industry (Wu et al., 2021). In a highly competitive environment, establishing a strong brand image is crucial for attracting consumers and gaining a competitive advantage.

- **Brand influence:** International brands hold significant recognition and wield considerable influence within the Chinese market. For example, Tesla, as a well-known international brand, has successfully attracted numerous consumers with its various advantages, such as high performance, remote sensing, and cutting-edge technology. Establishing a strong brand image is of great significance to Chinese NEV manufacturers (Pathak et al., 2022).
- **Channel construction and marketing strategy:** Competitors entice consumers by establishing a robust sales and service network, as well as implementing various marketing strategies (Zhang and Bai, 2017). Chinese NEV manufacturers should prioritize enhancing their channel infrastructure, intensifying marketing endeavors, and optimizing product sales and brand promotion to appeal to a wider audience.

4.2. Domestic and international market opportunities

The Chinese NEV industry is facing abundant domestic and international market opportunities. The Chinese government has been actively promoting the growth of the NEV sector by implementing a range of supportive policies and incentives, thereby creating favorable conditions for its rapid expansion (Lo, 2014). As global concerns regarding environmental preservation and sustainable development continue to intensify, NEVs have emerged as a promising clean energy alternative to traditional fuel-powered vehicles (Elavarasan et al., 2021). Consequently, they have garnered significant attention from the international market. The Chinese NEV industry has a clear competitive advantage with its technological research and development capabilities, strong production capabilities, and market expansion capabilities.

4.2.1. Domestic market opportunities

The domestic market for NEVs is characterized by a host of promising opportunities, including robust policy support, burgeoning market demand, continuous technological innovation, and expanding industrial chain development (Sun et al., 2014). These factors combined provide enormous growth potential and numerous business opportunities for the Chinese NEV market.

- **Green transformation policy support:** The Chinese government is committed to driving green transformation and actively supports the growth of NEVs by implementing a host of supportive policies (Mazzocco, 2020). These policies include car purchase subsidies (Dong et al., 2020), exemption from purchase tax (Ji et al., 2022), and free charging facilities (Liu et al., 2021), which play a key role in galvanizing market growth by providing robust support and stimulation.
- **Charging infrastructure construction:** The government is making substantial efforts to accelerate the construction of charging infrastructure, aiming to enhance the convenience of charging for NEV users. The continuous and steady increase in the number of public and private charging stations has created a more favorable environment for the widespread adoption and promotion of NEVs.
- **Accelerated urbanization process:** The acceleration of urbanization has created opportunities for the growth of NEVs. With its small size, flexibility, and zero emissions, NEVs are very suitable for urban environments (Kammen and Sunter, 2016). Therefore, they have enormous market potential in the field of urban transportation.

4.2.2. International market opportunities

NEVs have shown enormous development opportunities in the international market. Policy support, market demand, technological innovation, and international cooperation have jointly driven economic growth. These factors are important driving factors for the rapid development of NEVs on a global scale (Wang, 2022).

- **Policy support:** Governments worldwide are actively promoting the development of NEVs and implementing a series of policies to support their production and sales. Notably, the Chinese government has played a crucial role in providing substantial support through car purchase subsidies, charging infrastructure construction, and other areas (Wang et al., 2017). As a result, numerous international car manufacturers have been drawn to the Chinese market.
- **Market demand:** With the growing emphasis on environmental consciousness, there has been a consistent rise in consumer demand for eco-friendly, energy-saving, and low-carbon vehicles. NEVs offer advantages such as zero emissions and low energy consumption,

which perfectly align with this market demand. Consequently, NEVs possess significant potential within the international market.

- Technological innovation: The innovation in technology for NEVs remains ongoing, with the introduction of new materials (Weng et al., 2022), new energy sources (Shrivastava and Sharma, 2022), intelligent features (Wang et al., 2022), and even more technologies. This expansion provides wider possibilities for future NEV development. As an example, Tesla has been a leader in technological breakthroughs with its intelligent application in the electric vehicle industry.
- International cooperation: The field of NEVs necessitates cross-border collaboration, involving countries working together in technology research and development, market promotion, and the construction of charging infrastructure (Dzienis and McCaleb, 2022). By jointly promoting the development of NEVs, this collaboration becomes essential. For instance, China's partnership with countries like Germany and Japan in the NEV sector continues to strengthen, providing an opportunity for the global advancement of NEVs (Yuan et al., 2015).

5. Responses to challenges and opportunities

The rapid development of NEVs is inseparable from the continuous progress and innovation of technology. With the rapid development of technology, the introduction of various technologies and materials, and the promotion of reasonable policies, NEVs are moving towards a more efficient, reliable, and environmentally friendly future. The measures to address opportunities and challenges are shown in Fig. 4.

5.1. Technological innovation and development

Introducing new materials and technologies to enhance the battery capacity and energy density, thus addressing the issue of electric vehicle range anxiety (Zhou et al., 2023). Advancements in materials have the potential to enhance electrode materials, electrolytes, and separator materials, thereby increasing the energy density and charge transfer rate of batteries (Xiao et al., 2020; Liu et al., 2020).

The integration of intelligent driving technology into NEVs has made possible the development of self-driving vehicles, which has the potential to greatly improve transport efficiency and safety (Bayyou, 2019). Autonomous driving and intelligent traffic management are implemented through the use of visual sensors, LiDAR, and other devices for real-time environmental sensing, as well as high-precision maps and artificial intelligence algorithms (Yu et al., 2022). This integration offers promising opportunities to reduce the occurrence of traffic accidents while simultaneously improving vehicle operational efficiency and energy utilization.

To facilitate the widespread use of NEVs, it is crucial to develop efficient charging technologies and infrastructure, particularly focusing on fast charging and wireless charging. The implementation of fast charging technology significantly reduces charging time, which can enhance the convenience for EV users. In addition, wireless charging technology uses electromagnetic induction to charge batteries without physical plugging, providing users with a convenient charging experience and enhancing the attractiveness of electric vehicles (Shanmugam et al., 2022).

5.2. Policy support and encouragement

The implementation of preferential car purchase policies has a significant impact on boosting sales of NEVs. Purchase subsidies reduce the financial burden on consumers and it narrows the price gap between NEVs and conventional fuel vehicles. It plays a crucial role in easing the cost barrier (Breetz and Salon, 2018). This not only enhances the

economic feasibility of purchasing NEVs but also improves their overall viability in the market (She et al., 2017).

Financial support and subsidies are crucial for promoting the progress of the NEV industry chain, as well as promoting innovation and competitiveness of relevant enterprises and institutions (Wu et al., 2018). Financial subsidies can reduce the research and production costs of new energy vehicle manufacturers, thereby promoting technological progress and product development (Wu et al., 2020). Furthermore, extending loan support and offering tax incentives to the NEV industry chain also plays a significant role in fostering the sustainable development of related enterprises and institutions.

Strengthening policy support for the construction of charging facilities is meaningful for improving the convenience of charging infrastructure and accelerating the promotion of NEVs (Ji and Huang, 2018). The government should develop strategic plans and policies to incentivize relevant enterprises and institutions to expand the number and geographical coverage of charging stations, thereby removing barriers to charging and promoting the development of intelligent and efficient charging networks. Moreover, it is essential to promote the standardization and interoperability of charging devices, as this will enhance the availability of charging stations and ensure compatibility among different vehicles by improving the universal use of charging interfaces.

5.3. Environmental protection and sustainable development

The government is actively promoting the concept of green mobility, emphasizing the reduction in the use of fuel-based vehicles and the promotion of shared mobility modes. Shared mobility modes, including car-sharing and ride-sharing, are highly effective in reducing the number of privately owned cars and their usage frequency (Zhou et al., 2020). This approach not only enhances vehicle utilization, but also contributes to the reduction of traffic congestion and the minimization of environmental pollution.

Promoting the integration of NEVs and renewable energy can minimize reliance on non renewable fossil fuels. Renewable energy sources such as solar and wind energy have the characteristics of renewability and low carbon emissions, making them ideal choices for charging and supplying power to new energy vehicles (Kabeyi and Olanrewaju, 2022). The urgent task is to develop policy measures to encourage the utilization of renewable energy. This includes establishing renewable energy projects, such as photovoltaic and wind power, to provide reliable and clean energy for new energy vehicles.

The government places great emphasis on the significance of complete vehicle recycling, specifically highlighting the recycling of NEVs and the safe disposal and resource reuse of waste batteries. The utilization of retired batteries in alternative scenarios allows for the maximization of their remaining value while minimizing environmental pollution risks (Tian et al., 2024b). Moreover, scrap batteries can be recycled for valuable metal materials to produce new batteries (Miao et al., 2022). The government and industry can work together to establish a comprehensive waste battery recycling system. It involves providing guidance to businesses and individuals on the safe disposal of waste batteries or entrusting them to professional institutions for proper treatment.

5.4. Academic research and collaborative promotion

The academic community is encouraged to intensively examine the technical challenges and explore innovative directions within the field of NEVs. Additionally, scholars should strive to deepen academic exchanges and foster greater cooperation. The academic community can focus on fundamental technologies such as battery technology, drive systems, and intelligent transportation for NEVs through interdisciplinary research (Fu et al., 2020; Yu et al., 2020; He et al., 2022). The academic community can provide pioneering theoretical research and technological

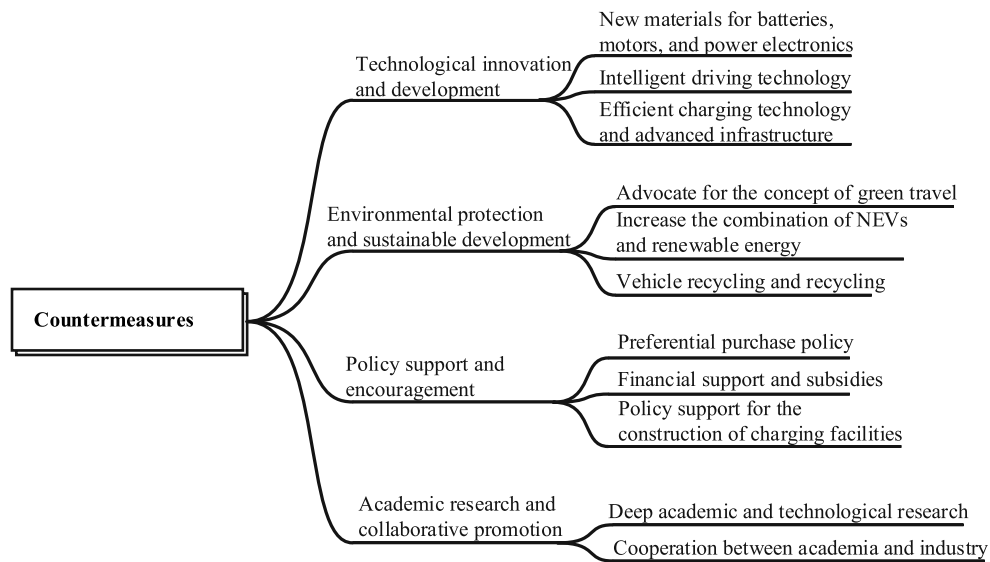


Fig. 4. Countermeasures for opportunities and challenges.

breakthroughs for the industry by dedicating themselves to these collaborative efforts.

The partnership between academia and industry is essential for translating scientific research into practical applications, driving technological innovation and industrial development in the field of NEVs. In order to achieve these goals, it is crucial to establish mechanisms for cooperation between industry and academia to promote joint development of new technologies and solutions to practical problems, while giving priority to the protection of intellectual property rights. Moreover, the promotion of technology transfer and the equitable sharing of intellectual property rights will further support the development of cooperation between academia and industry.

6. Conclusion

This work primarily analyses various NEV industry policies implemented by the Chinese government at the national and local levels, and summarizes the impact mechanisms of these policies on industrial development. This research holds significant importance in understanding the development of the Chinese NEV industry and its policy support.

The government has implemented various incentives such as vehicle purchase subsidies, exemption from vehicle taxes, and free license plates to strongly support and promote the development of the NEV industry. Additionally, the government has introduced product standards and energy efficiency standards for NEVs to drive technological innovation. Moreover, the government has released the New Energy Vehicle Industry Plan, which lists NEVs as strategic emerging industries and outlines their development plans and goals. These factors can be summarized as follows:

- (1) The car purchase subsidy policy has reduced consumers' purchasing costs, boosting the sales of NEVs.
- (2) The NEV quota management policy restricts the sales of traditional fuel vehicles, which promotes the use of NEVs.
- (3) The dual-point policy assesses both the energy consumption of enterprise product lines and the number of NEVs, with penalties for violations, driving companies to research and produce NEVs.
- (4) The construction of charging infrastructure has improved the convenience for users of NEVs.

From a macro perspective, the rapid development of the Chinese NEV industry also benefits from the progress of global science and technology.

On a global scale, the rapid development of the NEV industry has become an important link in promoting human social progress.

- (1) Global climate change is becoming increasingly severe, and environmental protection has become a common concern for the world. The large amount of greenhouse gases emitted by traditional fuel vehicles is one of the main driving factors of climate change. Therefore, promoting the development of NEV has become one of the key measures to mitigate climate change and achieve carbon neutrality goals.
- (2) Global fossil energy is facing an increasingly tense situation. NEV can reduce reliance on traditional energy and improve energy utilization efficiency. It is of great significance for global energy security and provides a new path for achieving sustainable development goals.
- (3) With global urbanization and demographic changes, consumers have an increasing demand for clean and convenient transportation. The environmental and low-carbon attributes of NEV are in line with the current consumer pursuit of quality living, and the market demand is constantly expanding. At the same time, the rise of the younger generation of consumers has also promoted the popularization of NEV, as they are more concerned about environmental protection and technology, and are willing to accept new technologies and products.
- (4) With global urbanization and demographic changes, consumers have an increasing demand for clean and convenient transportation. The popularization of NEV can effectively reduce exhaust emissions, reduce noise pollution, and promote the development of urban transportation towards smart and green directions.
- (5) The cooperation between Chinese car companies and international giants, the introduction of advanced technology, and cross-border research and development cooperation have made Chinese NEVs competitive in the international market, while also accelerating the innovation and development of the industry.

Overall, the rapid development of the Chinese NEV industry is driven by various factors such as policies, technology, and market, as well as the global trend of environmental protection, energy security, and industrial transformation. It is also an important link in promoting human social progress. The different countries and stakeholders should work together to strengthen cooperation, promote innovative development of the NEV

industry, and make positive contributions to building a sustainable, green, and intelligent future transportation system.

CRediT authorship contribution statement

Jaime Tian: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ping Wang:** Writing – review & editing, Formal analysis. **Daina Zhu:** Writing – review & editing, Formal analysis, Conceptualization.

Declaration of competing interest

There is no conflict of interest.

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